SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CHEMISTRY **9647/01**
JC2 Preliminary Examination **23 August 2011**
Paper 1 Multiple Choice **1 hour**

Additional Materials: Data Booklet
Optical Mark Sheet (OMS)

READ THESE INSTRUCTIONS FIRST

On the separate multiple choice OMS given, write your name, FIN/NRIC and class in the spaces provided.
Shade correctly your class and FIN/NRIC number.
Eg. If your NRIC is S9306660Z, shade **S9306660Z** for the item "index number".

There are **forty** questions in this paper. Answer **all** questions.
For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the one you consider correct and record your choice using a **soft pencil** on the separate OMS.
Each correct answer will score one mark.
A mark will not be deducted for a wrong answer.

You are advised to fill in the OMS as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Any rough working should be done in this question paper.

Section A

For each question there are four possible answers, **A, B, C** and **D**. Choose the one you consider to be correct.

1 A student added 10 cm^3 of 0.2 mol dm^{-3} KI (aq) to 10 cm^3 of 0.2 mol dm^{-3} $CuSO_4$ (aq) in a beaker and observed that a white precipitate in brown solution was obtained. She knew that if she were to add $S_2O_3^{2-}$ (aq) to the mixture, the brown solution will be decolourised.

Calculate the volume of 0.04 mol dm^{-3} $S_2O_3^{2-}$ (aq) she should add to the mixture in the beaker in order to completely decolourise the brown solution.

A 12.5 cm^3
B 15.0 cm^3
C 16.7 cm^3
D 25.0 cm^3

2 Which of the following contains the greatest amount (in mol) of particles?

A 600 g of I_2 (s)
B 50 cm^3 of H_2O (*l*)
C 50 dm^3 of HC*l* (g) at s.t.p.
D 550 cm^3 of 2.5 mol dm^{-3} CH_3COOH (aq)

3 An element, **E** can form a simple ion, $\mathbf{E}^{2+}$.

Which of the following is the electronic configuration of an atom of **E**?

A $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$
B $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$
C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$
D $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$

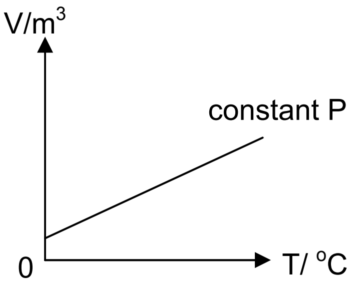
4 In 1996, Professor Hoffman and a group of scientists discovered a new element known as Ununbium, Uub. An atom of Uub can be formed by the fusion of a lead nucleus with the nucleus of an isotope of element **G** as follows:

$$^{208}_{82}\text{Pb} + \mathbf{G} \longrightarrow {}^{277}_{112}\text{Uub} + 1 \text{ neutron}$$

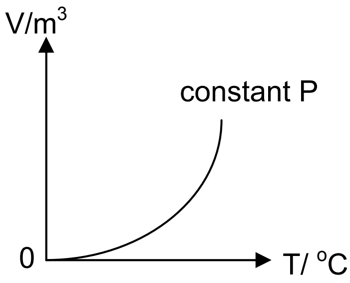
What is **G**?

A Cu
B Zn
C Ga
D Ge

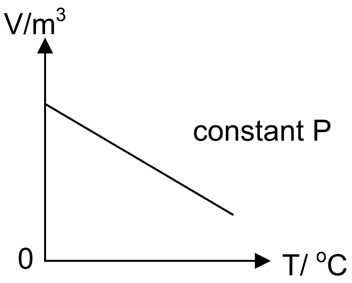
5 Which of the following graphs correctly describes the behaviour of a fixed mass of ideal gas?



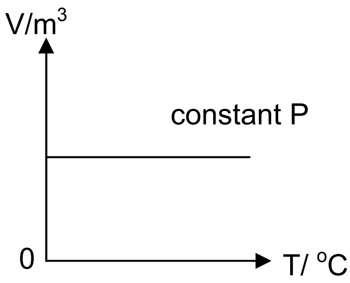
A



B



C



D

6 Which of the following consists of three compounds with different types of structure?

A BeO, SO_2, BH_3
B $BeCl_2$, Na_2O, SiO_2
C AlF_3, $MgCl_2$, BN
D $AlCl_3$, Cu_2O, $SiCl_4$

7 Ethyl ethanoate undergoes hydrolysis in water in the presence of dilute hydrochloric acid catalyst.

$$CH_3COOCH_2CH_3 + H_2O \xrightleftharpoons{HCl} CH_3COOH + CH_3CH_2OH$$

The rate of reaction was investigated by varying the concentrations of the ester and HC*l*.

Experiment	[ethyl ethanoate] / mol dm^{-3}	[HC*l*] / mol dm^{-3}	Initial rate / mol dm^{-3} min^{-1}
1	0.10	0.20	0.096
2	0.15	0.10	0.072
3	0.10	0.10	0.048

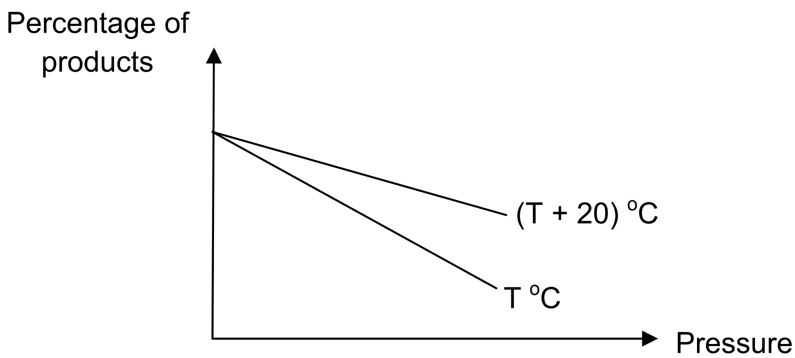
Which of the following statements about the above reaction is correct?

A Half-life for experiment 1 is 0.144 min.
B The rate equation of this reaction is rate = k[$CH_3COOCH_2CH_3$][H_2O].
C Tripling the concentration of both ethyl ethanoate and HC*l* increases the rate of reaction by a factor of 6
D If the half life of ethyl ethanoate in experiment 1 is 5.0 min, then the half life of ethyl ethanoate in experiment 3 will be 10.0 min.

8 When 2.4 g of Na_2CO_3 was added to 50 cm^3 of 1.0 mol dm^{-3} aqueous hydrochloric acid, the temperature of the solution rose by 3.2 ^{o}C What is the enthalpy change of neutralisation of this reaction? (Assume that 4.20 J is required to raise the temperature of 1 cm^3 of the solution by 1 K.)

A -1.34×10^4 J mol^{-1}
B -2.97×10^4 J mol^{-1}
C -3.11×10^4 J mol^{-1}
D -2.56×10^6 J mol^{-1}

9 The graph below shows how the percentage of products present at equilibrium varies with temperature and pressure.



Which one of the following reactions could the graph represent?

A $2Fe\,(s) + \frac{3}{2}O_2\,(g) \rightleftharpoons Fe_2O_3\,(s)$ $\Delta H = -822\ kJ\ mol^{-1}$

B $C\,(s) + \frac{1}{2}O_2\,(g) \rightleftharpoons CO\,(g)$ $\Delta H = -111\ kJ\ mol^{-1}$

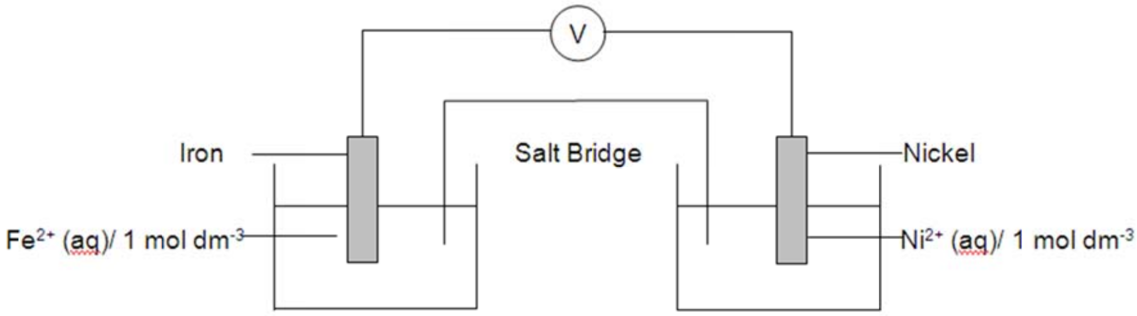
C $N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$ $\Delta H = +57\ kJ\ mol^{-1}$

D $CO\,(g) + Cl_2\,(g) \rightleftharpoons COCl_2\,(s)$ $\Delta H = +76\ kJ\ mol^{-1}$

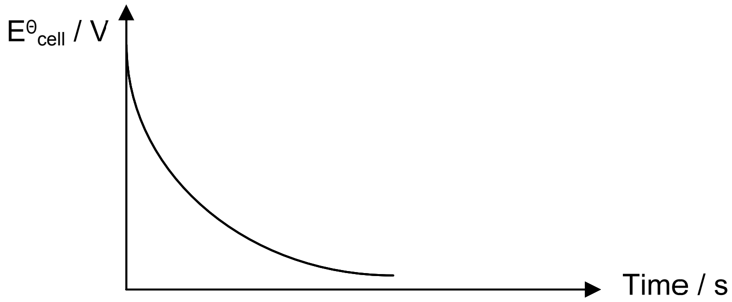
10 Given that the K_{sp} of $Mg(OH)_2$ is $1.5 \times 10^{-11}\ mol^3\ dm^{-9}$ and the K_b of aqueous $C_2H_5NH_2$ is $5.6 \times 10^{-4}\ mol\ dm^{-3}$, what is the solubility of $Mg(OH)_2$ in 1.0 $mol\ dm^{-3}$ aqueous $C_2H_5NH_2$?

A $1.64 \times 10^{-4}\ mol\ dm^{-3}$
B $4.78 \times 10^{-5}\ mol\ dm^{-3}$
C $2.68 \times 10^{-8}\ mol\ dm^{-3}$
D $6.34 \times 10^{-10}\ mol\ dm^{-3}$

11 An experiment is set up as shown below:



The $E^{\ominus}_{cell}$ of the cell was monitored as time progressed. When a change was made continuously to the set-up, the graph below was obtained.

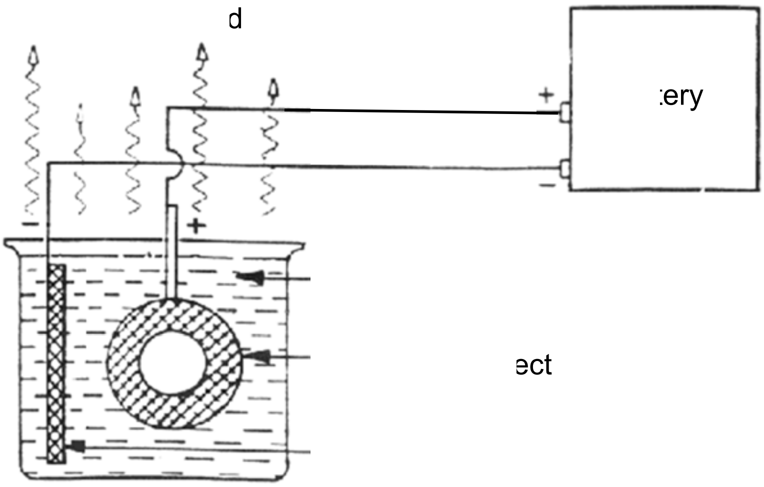


What was the continuous change made?

A Add nickel (II) chloride to the nickel half cell
B Add NaCN to the iron half cell
C Add water to the nickel half-cell
D Increase the surface area of iron immersed in the solution.

12 Aluminium alloys usually undergo a process known as anodisation to increase corrosion resistance and surface hardness. In this process, oxygen is formed and the gas reacts with the aluminium object to form a thick layer of aluminium oxide on the metal surface that protects the metal.

A possible set up for this process is shown as below:



Which of the following statements about the anodisation of aluminium is true?

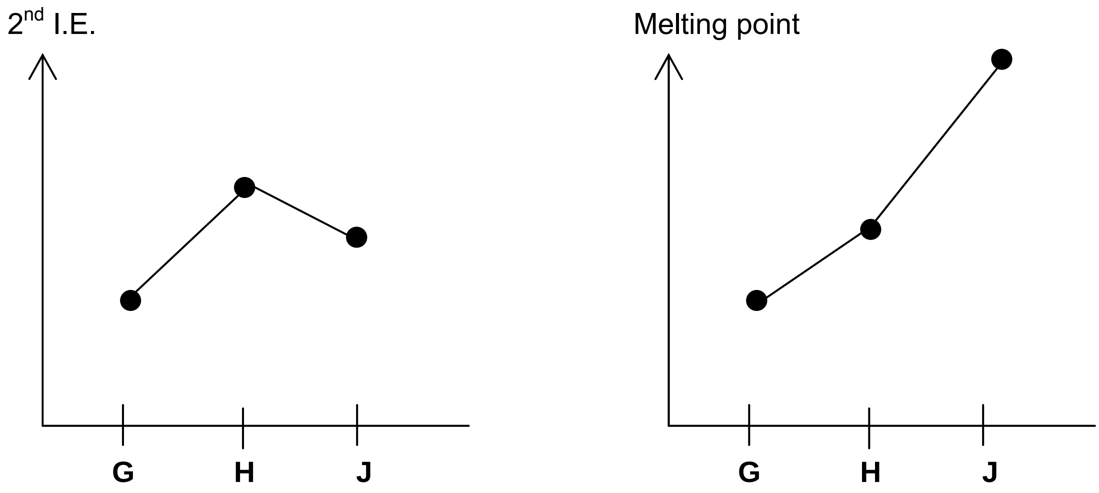
A Hydrogen gas and sulfur dioxide gas are liberated.

B Replacing the graphite electrode with copper will cause the reaction to cease.

C Anodisation is also possible when the aluminium object is connected to the negative terminal

D Water is oxidised at the anode to produce oxygen gas.

13 Consecutive elements **G**, **H** and **J** are in Period 3 of the Periodic Table. The trend of their second ionisation energies and melting points are shown in the graphs below.



What could be the identities of **G**, **H** and **J**?

A Na, Mg, A*l*
B Mg, A*l*, Si
C Si, P, S
D S, C*l*, Ar

14 Which of the follow properties of beryllium and its compounds is incorrect?

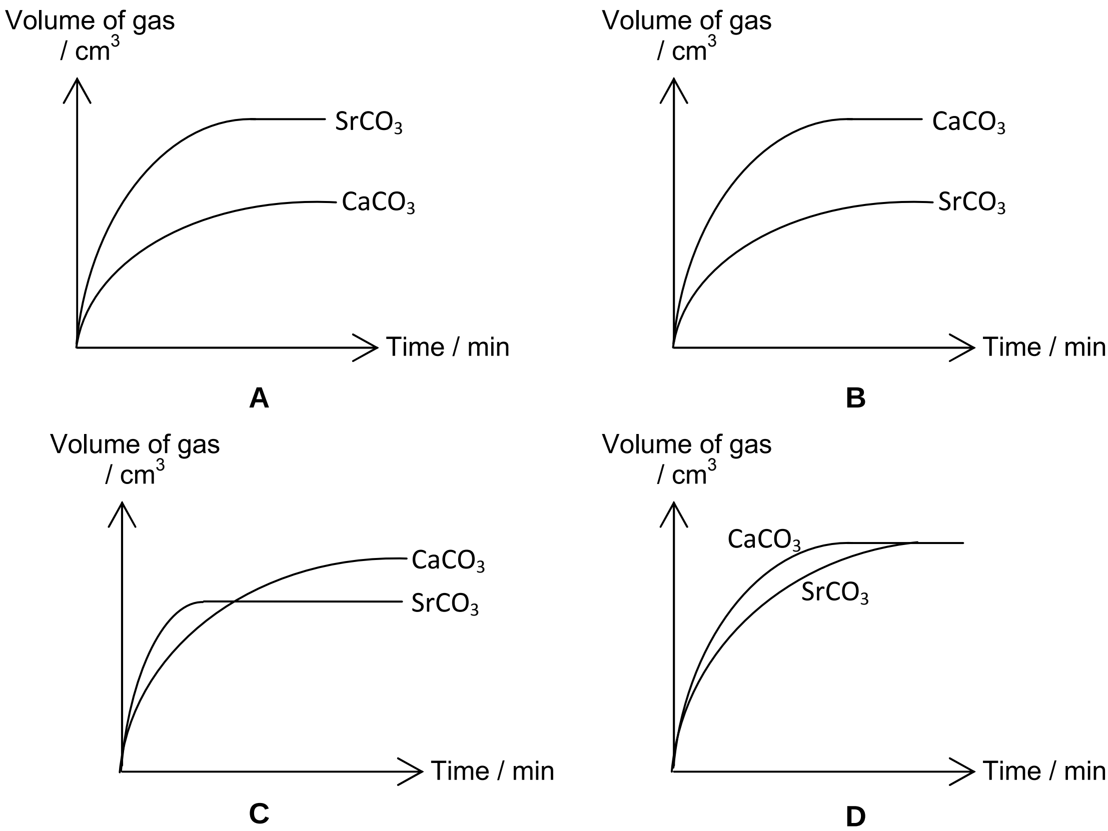
A Beryllium does not react with water.
B Beryllium oxide undergoes neutralisation with both acids and bases.
C Beryllium chloride reacts with ammonia in a 1:2 ratio.
D Beryllium chloride is insoluble in water

15 When concentrated sulfuric acid was added to solid potassium chloride, white fumes of hydrogen chloride gas was observed. The same observation was made when concentrated sulfuric acid was added to solid potassium iodide, but significantly less white fumes was observed.

Which of the following explains the lower yield of hydrogen iodide as compared to hydrogen chloride?

A Iodine is less reactive than chlorine.
B Iodine is a weaker base than chlorine.
C Hydrogen iodide is less volatile than hydrogen chloride.
D Hydrogen iodide is more easily oxdised than hydrogen chloride.

16 Which of the graphs below show the variation in the volume of carbon dioxide gas collected when 1 mol of strontium carbonate and 1 mol of calcium carbonate are heated strongly?

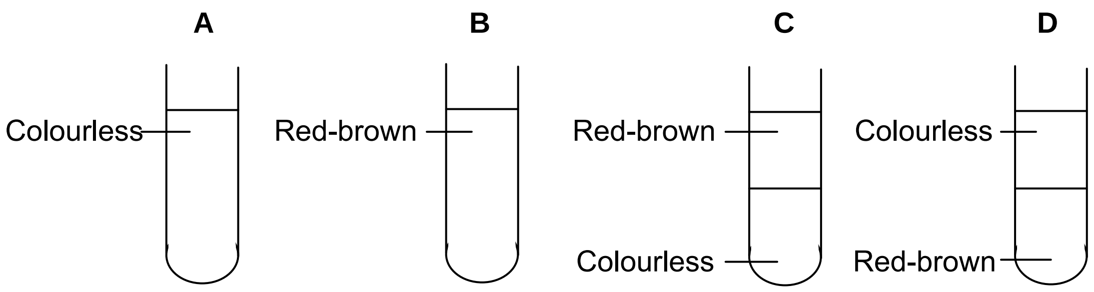


17 Astatine is below iodine in Group VII. Which of the following properties is consistent with its position in the Periodic Table?

A Silver astatide is soluble in dilute ammonia.
B Astatine is a liquid at room temperature and pressure.
C Astatine reacts with aqueous Fe (II) to form Fe (III).
D Hydrogen astatide decomposes at a lower temperature than hydrogen iodide.

18 Aqueous chlorine is added to aqueous sodium bromide and the mixture was shaken with an equal amount of trichloromethane.

Given that the density of trichloromethane is 1.48 g cm^{-3}, which one of the following observations can be seen?



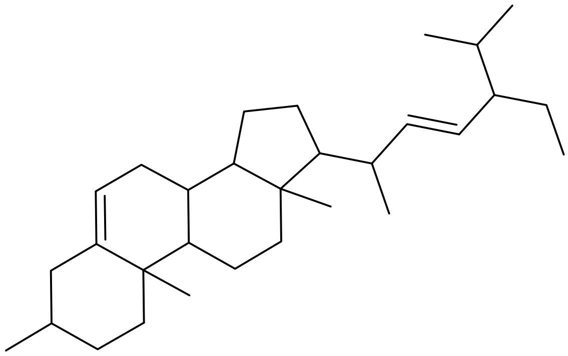
19 Below is a scheme of reaction showing the chemistry of some compounds of transition metals in aqueous solution.

$$\text{Cu (s)} \xrightarrow[\quad]{\text{Excess } AgNO_3\text{ (aq)}} \text{Blue solution} \xrightarrow{\text{Excess KC}l\text{ (aq)}} \text{White precipitate}$$

What are the possible identities of the blue solution and the white precipitate?

	Blue solution	White precipitate
A	$CuNO_3$	CuC*l*
B	$CuNO_3$	AgC*l*
C	$Cu(NO_3)_2$	CuC*l*
D	$Cu(NO_3)_2$	AgC*l*

20 Stigmasterol is an unsaturated plant sterol occurring in the plant fats of soybean and rape seed.



How many stereoisomers does stigmasterol have?

A 2^9
B 2^{10}
C 2^{11}
D 2^{12}

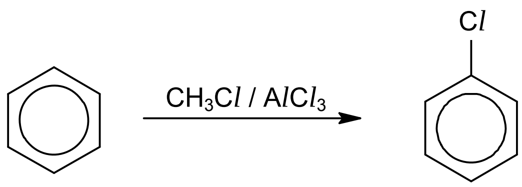
21 In which of the following reactions is the reactive carbon sp^3 hybridised in the reactant and sp^2 hybridised in the product?

A $CH_3CHCHCH_3 \xrightarrow{\text{Br}_2 \text{ in CC}l_4\text{, in the dark}} CH_3CHBrCHBrCH_3$

B $CH_3COCH_3 \xrightarrow{\text{LiA}l\text{H}_4 \text{ in dry ether, r.t.p.}} (CH_3)_2CHOH$

C $CH_3CH_2OH \xrightarrow{\text{Excess conc. H}_2\text{SO}_4\text{, 170}^0\text{C}} CH_2CH_2$

D



22 Which one of the following processes is a propagation step in the chain reaction between CH_2Cl_2 and Cl_2 when irradiated with light?

A $Cl_2 \longrightarrow 2\,Cl\bullet$

B $CH_2Cl_2 + Cl_2 \longrightarrow CHCl_3 + HCl$

C $\bullet CCl_3 + Cl_2 \longrightarrow CCl_4 + Cl\bullet$

D $CHCl_3 + Cl\bullet \longrightarrow CCl_4 + H\bullet$

23 In the preparation of ethene, ethanol was added to a drop of heated reagent **L**. The impure ethene was washed by being bubbled through a solution of **M** before collection. What are the reagents **L** and **M** likely to be?

	Reagent **L**	Reagent **M**
A	concentrated H_2SO_4	ethanolic NaOH
B	concentrated H_2SO_4	dilute NaOH
C	ethanolic NaOH	concentrated H_2SO_4
D	dilute NaOH	concentrated H_2SO_4

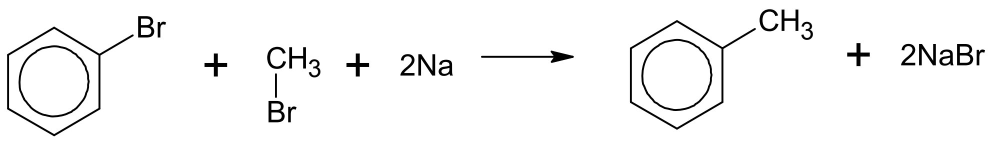
24 In the Wurtz reaction, two halogenoalkanes react with sodium metal to form a new carbon-carbon bond, resulting in the formation of a new alkane:

$$R{-}X + R'{-}X + 2Na \longrightarrow R{-}R' + 2NaX$$

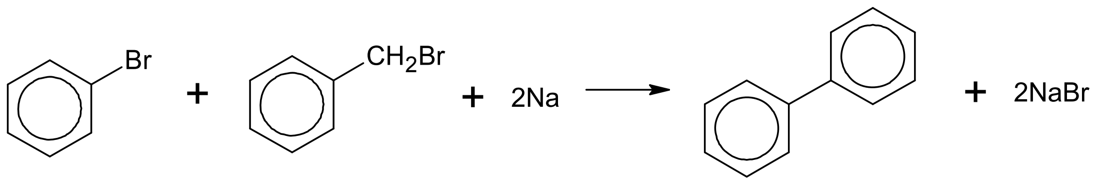
Which of the following does not show the correct product when the stated reactants are reacted together in a Wurtz reaction?

A $CH_3Br + C_2H_5I + 2Na \longrightarrow CH_3CH_2CH_3 + NaBr + NaI$

B



C



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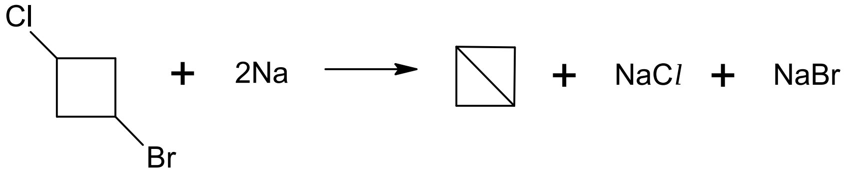
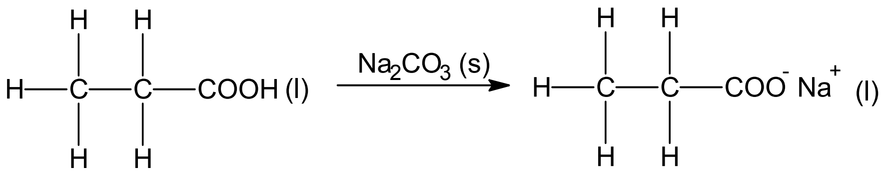
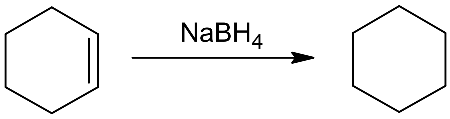
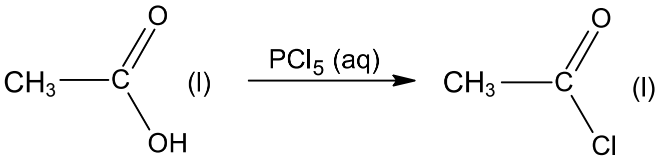
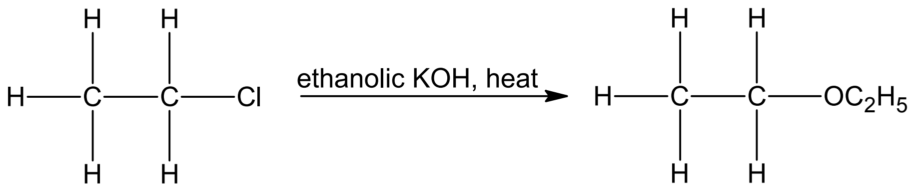


25 Which of the following shows the correct reagents and conditions to produce the desired product?

A 

B 

C 

D 

26 Compound **Q** was refluxed with aqueous sodium hydroxide and the resulting mixture was then distilled. The distillate gave a positive tri-iodomethane test. The residue in the distillation flask, after acidification, gave a white precipitate.

Which of these could be **Q**?

A $CH_3CH_2COOCH_2CH_3$
B $C_6H_5COOCH_3$
C $CH_3CH_2OCOC_6H_5$
D $CH_3CONHC_6H_5$

27 Which of the following produces a compound with a chiral carbon centre on reaction with hydrogen cyanide?

A CH_3CHO
B $CH_3CH_2COCH_2CH_3$
C $CH_3CO_2CH_3$
D HCHO

28 The Russian composer Borodin, was also a research chemist. He discovered a reaction in which two ethanal molecules combine to form a compound commonly known as an aldol (reaction I). The aldol formed can then produce another compound on heating (reaction II).

reaction I $2CH_3CHO \longrightarrow CH_3CH(OH)CH_2CHO$

reaction II $CH_3CH(OH)CH_2CHO \longrightarrow CH_3CH{=}CHCHO + H_2O$

Which of the following best describes reactions I and II?

	I	II
A	addition	elimination
B	addition	reduction
C	elimination	reduction
D	substitution	elimination

29 Compounds **W**, **X** and **Y** react with sodium, but only one of them reacts with aqueous alkaline iodine. Which of the following combinations is likely to be **W**, **X** and **Y** respectively?

A C_6H_5OH, CH_3COOH, $(CH_3)_3COH$

B $HOCH_2CH_2OH$, $HOCH(CHI_2)CH_2COOH$, CH_3COCH_2I

C CH_3COOH, $(CH_3)_3COH$, CH_3COCH_2OH

D CH_3COOH, CH_3COCHI_2, CH_3COOCH_2OH

30 In the study of a polypeptide structure of **Z**, it was digested using two different enzymes. The fragments obtained were then separated using electrophoresis. Analysis of the fragments from each digestion gave the following results:

Fragments using first enzyme:
tyr-leu-leu
tyr-ala
gly-asp-pro
asp-pro

Fragments using second enzyme:
leu-tyr
asp-pro-gly
ala
asp-pro-tyr-leu

Deduce the possible sequence of **Z**.

A asp-pro-tyr-leu-ala-leu-tyr-asp-pro-gly
B asp-pro-gly-asp-pro-tyr-leu-leu-tyr-ala
C gly-asp-pro-asp-pro-tyr-ala-tyr-leu-leu
D ala-asp-pro-gly-asp-pro-tyr-leu-leu-tyr

Section B

For **questions 31-40**, one or more of the numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct. The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, **2** and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is to be used as a correct response.

31 Two colourless liquids were mixed together in a beaker and left to stand. The mixture separated into two distinct layers after standing for an hour.

Assuming that the liquids did not undergo any reaction with each other, which pair(s) of liquids, when mixed, will produce the above observation?

1 Ethanal and water
2 Cyclohexanol and methanol
3 Ethanol and tetrachloromethane

32 A reversible reaction, **A** (aq) + **B** (aq) $\rightleftharpoons$ **C** (aq) is catalysed by an aqueous solution of M^{2+} ions.

Which of the following statement(s) about this system is/are correct?

1 The catalyst alters the mechanism of the reaction.
2 The catalyst reduces the activation energy for both the forward and backward reaction.
3 The catalyst alters the composition of the equilibrium mixture.

33 Sulfuryl chloride, SO_2Cl_2, is often used as a source of Cl_2 for various organic reactions. Sulfuryl chloride is also a highly reactive gaseous compound which decomposes endothermically when heated in the following manner:

$$SO_2Cl_2\,(g) \rightleftharpoons SO_2\,(g) + Cl_2\,(g)$$

A 7:2 mole ratio of SO_2Cl_2 and Cl_2 was placed in an evacuated vessel at 375 K and 6 atm. After 10 minutes, the mixture reached equilibrium and the partial pressure of SO_2 was found to be 0.625 atm.

Based on the above data, which of the following statement(s) is/are correct?

1 The K_p value is 0.303 atm.
2 The K_p value will decrease when total pressure increases.
3 The partial pressure of SO_2Cl_2 will remain constant when temperature decreases.

34 Consider the reaction between sulfur dioxide and oxygen to form sulfur trioxide:

$$2SO_2 + O_2 \longrightarrow 2SO_3$$

Given that $\Delta H = -199$ kJ mol^{-1}, and $\Delta S = -190$ J K^{-1} mol^{-1} for the above reaction, which of the following statement(s) is/are correct for the reaction?

1 The reaction is spontaneous under standard conditions.
2 As temperature increases, the reaction becomes more spontaneous.
3 In the formation of sulfur trioxide, the system becomes more disordered.

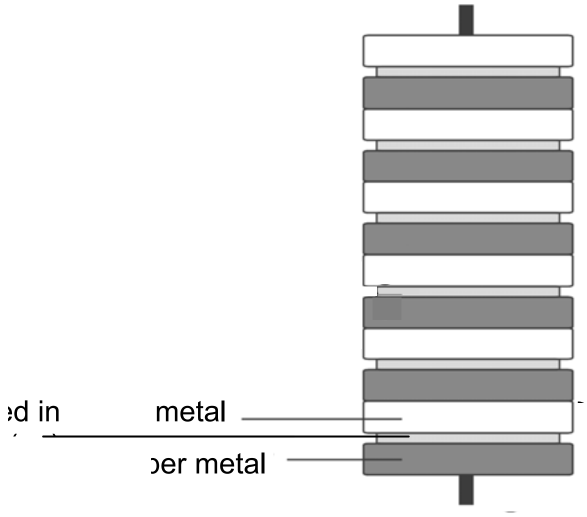
35 Hydrazoic acid, HN_3, is an acid with pK_a value of 4.72.

Which of the following statement(s) about a 25.0 cm^3 sample of 0.100 mol dm^{-3} HN_3 (aq) is/are correct?

1 The concentration of N_3^- in the sample is 1.38×10^{-3} mol dm^{-3}.
2 A buffer solution is formed when 12.50 cm^3 of 0.100 mol dm^{-3} of NH_3 (aq) is added to the sample.
3 When titrated with NaOH (aq), the equivalence pH is above 7.

36 In 1800, an Italian physicist Alessandro Volta invented the first battery – the voltaic pile.

A diagram of the voltaic pile is shown below:



Which of the following statement(s) is/are correct?

1 The overall equation of the reaction occurring in the cell is $Cu^{2+} + Zn \rightarrow Zn^{2+} + Cu$.
2 Zinc is the anode, while copper is the cathode.
3 The e.m.f. of the voltaic pile can be increased by stacking more elements

37 When drops of NaOH (aq) were added to a green solution of $Cr(NO_3)_3$ (aq), a grey-green precipitate was formed. The precipitate dissolved when excess NaOH(aq) was added, forming a dark green solution. Subsequent additions of liquid ammonia caused the solution to turn violet.

According to the information given above, which of the following statement(s) is/are incorrect?

1 OH^- acted as a ligand in the first addition of drops of NaOH (aq).
2 NH_3 is a stronger field ligand as compared to OH^-.
3 The $[Cr(H_2O)_6]^{3+}$ complex is the most unstable of all the complexes involved.

38 Chloroethane reacts with excess alcoholic ammonia when heated to form a quaternary ammonium salt in four steps.

$$C_2H_5Cl \xrightarrow{NH_3} C_2H_5NH_2 \xrightarrow{C_2H_5Cl} (C_2H_5)_2NH \xrightarrow{C_2H_5Cl} (C_2H_5)_3N \xrightarrow{C_2H_5Cl} (C_2H_5)_4N^+$$

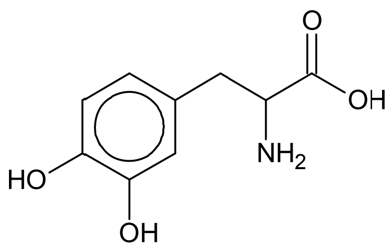
Which of the following statement(s) about the reaction is/are correct?

1 In each step, the attacking nucleophile is more nucleophilic than that in the previous step.

2 In each step, the reaction is slower than the previous step due to increased steric hindrance.

3 In each step, an intermediate with a sp^2 hybridised carbon atom is formed.

39 Dopamine is a neurotransmitter found in many animals, including vertebrates and invertebrates. The structure of dopamine is shown below:



Which of the following statement(s) is/are correct?

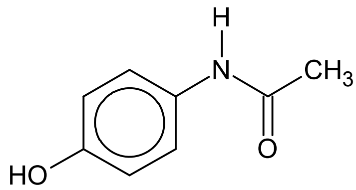
1 1 mol of dopamine reacts with 3 mol of ethanoyl chloride.

2 1 mol of dopamine reacts with 1 mol of HBr (aq).

3 1 mol of dopamine reacts with 1 mol of Na_2CO_3.

40 Paracetamol is commonly used for the relief of headaches, fever and pains. It is a major ingredient for cold and flu remedies.

The structure of Parcetamol is shown below:



Which of the following statement(s) about Paracetamol is/are correct?

1 On addition of concentrated nitric acid, a white precipitate will be formed.

2 A pale yellow precipitate is formed when a solution of warm alkaline aqueous iodine solution is added.

3 It reacts with hot aqueous sodium hydroxide to produce a gas which turns damp red litmus paper blue